

Numerical Shape Optimization of a Vortex Flowmeter Using Numerical Methods and PyLBM

Section A1 — Group 6

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Abstract

Vortex flowmeters estimate flow velocity from the frequency of alternating vortices produced behind a bluff body. The generator shape affects both the strength of the measurable signal and the pressure loss introduced into the pipe. This project presents a fully Python-based numerical workflow for studying the curved side radius of a paper-inspired triangular vortex generator. PyLBM was used to solve a two-dimensional D2Q9 central-moment multiple-relaxation-time lattice Boltzmann model. Five side radii between 20 and 40 mm were simulated. Wake-pressure samples were processed using least-squares detrending, peak detection and the fast Fourier transform. Trapezoidal and Simpson rules were used to calculate mean pressure loss, while numerical gradients were used to obtain vorticity. A normalized objective rewarded shedding frequency and signal amplitude and penalized pressure loss and signal asymmetry. Lagrange interpolation, Newton interpolation, a natural cubic spline and QR least-squares fitting were compared. Golden-Section Search on the QR cubic predicted a candidate radius of 39.189 mm. A fresh simulation at this radius produced an objective of 0.2337 and passed stability, low-Mach, mass-drift, stationarity, frequency-agreement and integration checks. The result is reported as the best verified candidate inside the tested interval rather than a proven global optimum.

Keywords

vortex flowmeter, lattice Boltzmann method, PyLBM, shape optimization, FFT, numerical integration, interpolation

1 Introduction and Base Paper Review

1.1 Background

A flowmeter measures the amount of fluid passing through a pipe. For a circular pipe, volumetric flow rate is

$$Q = UA = U \frac{\pi D_p^2}{4}, \quad (1)$$

where Q is volumetric flow rate, U is average velocity, A is pipe area and D_p is pipe diameter.

A vortex flowmeter places a bluff body, called a vortex generator, in the flow. The fluid separates from the body and forms alternating vortices downstream. This repeating wake is known as a von Kármán vortex street. The shedding frequency is related to velocity by the Strouhal number,

$$St = \frac{fD}{U}, \quad (2)$$

where f is shedding frequency and D is a characteristic generator width. If D and St are known, measuring f provides an estimate of U .

Generator shape is important because it controls flow separation, vortex strength, wake symmetry and pressure loss. A strong signal is desirable for measurement, but excessive pressure loss increases the energy required to move fluid through the pipe. Shape design is therefore a balance between measurement quality and hydraulic resistance.

1.2 Review of the Base Paper

The selected base paper is “Shape Optimization of the Triangular Vortex Flowmeter Based on the LBM Method” by Sun et al. [4]. The authors used the OpenLB framework with a three-dimensional D3Q19 MRT-LBM large-eddy simulation model. They optimized three geometric features: the incoming-flow surface, the concave side and the downstream end. Their selected geometry had a 180° incoming surface, a 25 mm side curvature radius and a 4 mm end fillet. The study evaluated shedding frequency, lift force per unit area, vortex-street symmetry and pressure loss. It reported increased shedding frequency and reduced pressure loss compared with the original triangular generator [4].

The paper’s model is more detailed than the present course implementation. It uses a three-dimensional LBM–LES model at high Reynolds numbers, whereas this project uses a reduced two-dimensional PyLBM model at $Re = 500$. Therefore, the present numerical values are not expected to reproduce the paper exactly.

1.3 Objective and Scope of This Project

To keep the experiment computationally manageable, the paper-selected incoming angle and end fillet were fixed at 180° and 4 mm. The side radius R was treated as the design variable. The study aimed to:

- (1) implement a reproducible Python/PyLBM simulation;
- (2) extract vortex frequency, signal amplitude, pressure loss and symmetry;
- (3) demonstrate numerical integration, interpolation, least-squares fitting and optimization;
- (4) predict a promising radius and check it with a new simulation.

The public source-code repository is available at <https://github.com/FAHIM-ISHTIAK/CSE-402-Project>.

2 Methodology

2.1 Parameterized Generator Geometry

The numerical body is constructed from a solid triangle, two circular fluid cuts and a solid circular cap. The triangle supplies the basic bluff-body shape. The fluid circles remove material from the upper and lower sides to create concave arcs. The cap rounds the downstream tip. The side radius R changes both curved cuts. Generator height, length, incoming angle and fillet radius remain fixed.

This representation follows the general paper-inspired geometry while allowing R to be changed automatically before each simulation. The tested values were 20, 25, 30, 35 and 40 mm.

2.2 Lattice Boltzmann Model

PyLBM [3] performs the collision, streaming and boundary operations. The general LBM update is

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) + \Omega_i, \quad (3)$$

where f_i is the distribution population, $\mathbf{c}_i$ is a lattice direction and Ω_i is the collision operator. The project uses a D2Q9 Geier central-moment MRT scheme. Macroscopic variables are recovered by

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{c}_i. \quad (4)$$

Pressure is reconstructed using the weakly compressible LBM relation

$$p = c_s^2 \rho, \quad (5)$$

and converted to physical units using the configured density and velocity scales.

2.3 Pressure Probes and Wake Signal

Four virtual probes are used. Upstream and downstream pressures are averaged over vertical bands before and after the body. Two fixed 3×3 -cell wake regions are placed symmetrically above and below the wake centerline.

Pressure loss is

$$\Delta p(t) = P_{upstream}(t) - P_{downstream}(t). \quad (6)$$

The vortex signal is

$$s(t) = P_{upper}(t) - P_{lower}(t). \quad (7)$$

Alternating vortices create pressure disturbances above and below the centreline. Their difference produces an oscillating signal suitable for frequency analysis.

2.4 Signal Processing and Frequency

A constant and linear trend are fitted to $s(t)$ using least squares and removed. Peak detection then identifies local maxima, rejects inconsistent periods and computes

$$f_{peak} = \frac{1}{\text{mean}(T_k)}. \quad (8)$$

An independent frequency estimate is obtained by applying a Hann window and a real FFT. Frequencies too low to complete two cycles inside the observation window are ignored. The highest remaining spectral amplitude defines f_{FFT} . Frequency resolution is

$$\Delta f = \frac{1}{T}, \quad (9)$$

where T is the sampled duration. A case is accepted only when the peak and FFT frequencies agree within the implemented tolerance.

2.5 Numerical Integration

For sampled pressure-loss points (t_j, y_j) , the trapezoidal integral is

$$I_{trap} = \sum_{j=0}^{N-2} \frac{y_j + y_{j+1}}{2} (t_{j+1} - t_j). \quad (10)$$

For uniformly spaced data with an even number of intervals, Simpson's rule is

$$I_{Simp} = \frac{h}{3} \left[y_0 + y_{N-1} + 4 \sum y_{odd} + 2 \sum y_{even} \right]. \quad (11)$$

Each integral is divided by T to obtain mean pressure loss. Their absolute difference is used as an integration-consistency indicator.

2.6 Vorticity and Flow Fields

Velocity magnitude is

$$|\mathbf{u}| = \sqrt{u_x^2 + u_y^2}. \quad (12)$$

The two-dimensional out-of-plane vorticity is

$$\omega_z = \frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y}. \quad (13)$$

The derivatives are approximated using second-order numerical array gradients.

2.7 Objective Function

For each radius, the code obtains shedding frequency $f(R)$, signal amplitude $A(R)$, mean pressure loss $\overline{\Delta p}(R)$ and symmetry deviation $S(R)$. These quantities have different units, so they are normalized using the first sweep case as a baseline. The objective is

$$J(R) = 0.35 \frac{f(R)}{f_0} + 0.20 \frac{A(R)}{A_0} - 0.30 \frac{\overline{\Delta p}(R)}{\overline{\Delta p}_0} - 0.15 \frac{S(R)}{S_0}. \quad (14)$$

Frequency and amplitude are rewarded. Pressure loss and asymmetry are penalized. The weights express project priorities; they are not universal physical constants. The optimization problem is

$$R^* = \arg \max_{R \in [20, 40]} J(R). \quad (15)$$

2.8 Interpolation, Fitting and Search

The five direct points (R_i, J_i) are studied using four approximations:

- Lagrange polynomial interpolation;
- Newton divided-difference interpolation;
- natural cubic spline interpolation;
- degree-three QR least-squares fitting.

Lagrange and Newton interpolation represent the same degree-four polynomial in different forms. The spline uses four local cubic sections. The QR model fits

$$\hat{J}(R) = a_0 + a_1 R + a_2 R^2 + a_3 R^3 \quad (16)$$

without forcing the curve through every potentially noisy simulation point. The current implementation searches this QR cubic using Golden-Section Search and Newton's optimization method. The 401 curve values saved for plotting are formula evaluations, not extra LBM simulations.

3 Implementation and Experiments

3.1 Software Structure

The implementation is written in Python. NumPy supports array calculations [1], Matplotlib creates the figures [2], and PyLBM supplies the flow-solver framework [3]. Important modules include:

- `config.py`: parameters and physical/lattice conversions;
- `geometry.py`: parameterized solid body;
- `lbm.py`: PyLBM configuration, execution and probes;
- `postprocess.py`: frequency, pressure and vorticity analysis;
- `numerics.py`: integration, interpolation, fitting and searches;
- `optimization.py`: objective and surrogate construction;
- `pipeline.py`: sweeps and verification;
- `plots.py`: reproducible figures.

3.2 Simulation Setup

Table 1 lists the main settings. The inlet lattice velocity was kept low to support the weakly compressible LBM assumption.

The sampled sequence is 30,000, 30,005, . . . , 54,000. Therefore,

$$N = \frac{54000 - 30000}{5} + 1 = 4801 \quad (17)$$

points and 4,800 intervals are used. The trapezoidal rule uses all 4,801 points. Simpson's rule groups the 4,800 intervals into 2,400 pairs.

3.3 Experimental Procedure

The experiment followed these steps:

- (1) load and validate the configuration;
- (2) build a new geometry for one radius;
- (3) run PyLBM for 54,000 steps;
- (4) discard the first 30,000 transient steps;
- (5) record pressure signals every five steps;

Table 1: Course simulation settings.

Parameter	Value
Grid	360×96
Time steps	54,000
Sampling begins	Step 30,000
Sample interval	5 steps
Reynolds number	500
Inlet lattice velocity	0.03
Physical velocity	10.21 m/s
Pipe diameter	0.09718 m
Characteristic width	0.028 m
Density	27.1721 kg/m ³
Blockage ratio	0.30

- (6) compute frequency, pressure loss, vorticity and quality metrics;
- (7) repeat for all five radii;
- (8) calculate the normalized objective;
- (9) construct the interpolation and fitted curves;
- (10) search the QR cubic;
- (11) run a fresh simulation at the Golden-Section candidate.

Cases were rejected if they failed stability, low-Mach, mass-drift, stationarity or frequency-reliability checks. Because the domain has open inlet and outlet boundaries, total domain mass may drift slightly; the implemented acceptance limit is 2%.

4 Results and Discussion

4.1 Radius Sweep

Table 2 reports the direct PyLBM results. No single raw metric defines the best geometry. For example, the 40 mm case has the highest signal amplitude and lowest listed pressure loss, but symmetry and frequency also affect the objective.

4.2 Interpolation and Optimization

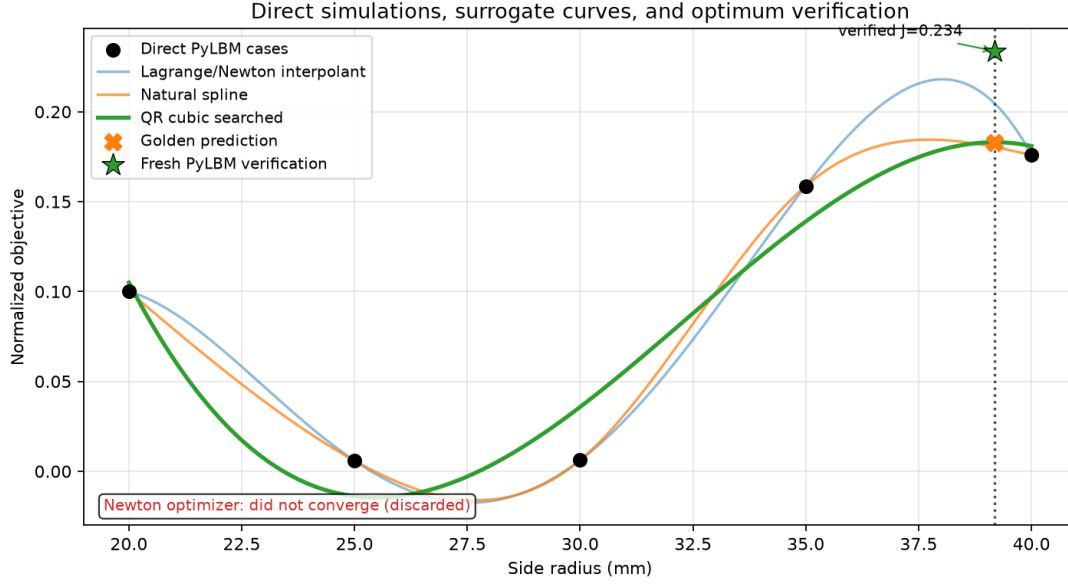
Figure 1 compares direct simulations with the surrogate curves. The fitted methods place the promising region near 38–40 mm. Lagrange/Newton interpolation predicts approximately 38.05 mm, the natural spline predicts approximately 37.75 mm, and the QR cubic predicts approximately 39.19 mm.

Golden-Section Search converged to 39.189 mm in 41 iterations with a predicted objective of 0.1830. Newton's method moved from 30 mm to a stationary point at 25.473 mm. At that point, the first derivative was approximately zero, but the second derivative was positive. It was therefore a local minimum, not a maximum, and the code correctly marked the result as not converged to a valid maximum.

The QR residual norm was 0.04125. This non-zero value confirms that the cubic does not pass through every direct result. A spline may represent local changes better, while the QR fit may suppress numerical variation. The disagreement among the curves is a reason to report an optimum region and perform direct verification rather than trusting a surrogate alone.

Table 2: Direct PyLBM sweep results.

R (mm)	f_{peak} (Hz)	f_{FFT} (Hz)	St	A (Pa)	$\overline{\Delta p}$ (Pa)	J
20	86.29	84.03	0.2367	10,051	4,075	0.1000
25	90.64	98.04	0.2486	9,484	4,086	0.0059
30	87.69	84.03	0.2405	9,327	3,995	0.0062
35	91.40	84.03	0.2507	7,988	4,060	0.1587
40	92.32	84.03	0.2532	10,801	3,991	0.1759

**Figure 1: Direct radius simulations, response approximations and candidate verification. The continuous curves are formula evaluations rather than additional LBM simulations.**

4.3 Verified Wake Signal and Spectrum

Figure 2 shows the fresh 39.189 mm case. The upper trace is $P_{upper} - P_{lower}$ and represents alternating wake pressure. The lower trace is $P_{upstream} - P_{downstream}$ and represents instantaneous pressure loss. The sampling window passed the pressure and signal-RMS stationarity checks.

Figure 3 contains the FFT of the detrended, Hann-windowed wake signal. The main non-zero peak is the dominant shedding frequency. Smaller components represent harmonics, waveform shape, transient content or numerical noise. The FFT estimate of 84.033 Hz differed from the peak-to-peak estimate of 84.856 Hz by only 0.98%, so frequency reliability passed.

4.4 Velocity and Vorticity Fields

Figure 4 shows the final saved spatial snapshot. The velocity panel displays acceleration around the body and the disturbed downstream region. The vorticity panel shows alternating positive and negative rotation, providing spatial evidence of a von Kármán vortex street. This image is an instantaneous snapshot, not a time average.

Table 3: Verification result at $R = 39.189$ mm.

Metric	Value
Peak frequency	84.856 Hz
FFT frequency	84.033 Hz
Frequency difference	0.98%
Strouhal number	0.2327
Signal amplitude	10,532.71 Pa
Symmetry deviation	0.02336
Trapezoidal pressure loss	3,931.359 Pa
Simpson pressure loss	3,931.358 Pa
Integration difference	0.000671 Pa
Mass drift	1.084%
Maximum lattice Mach	0.1698
Verified direct objective	0.2337

4.5 Verification Metrics

Table 3 summarizes the fresh candidate simulation. All implemented quality checks passed.

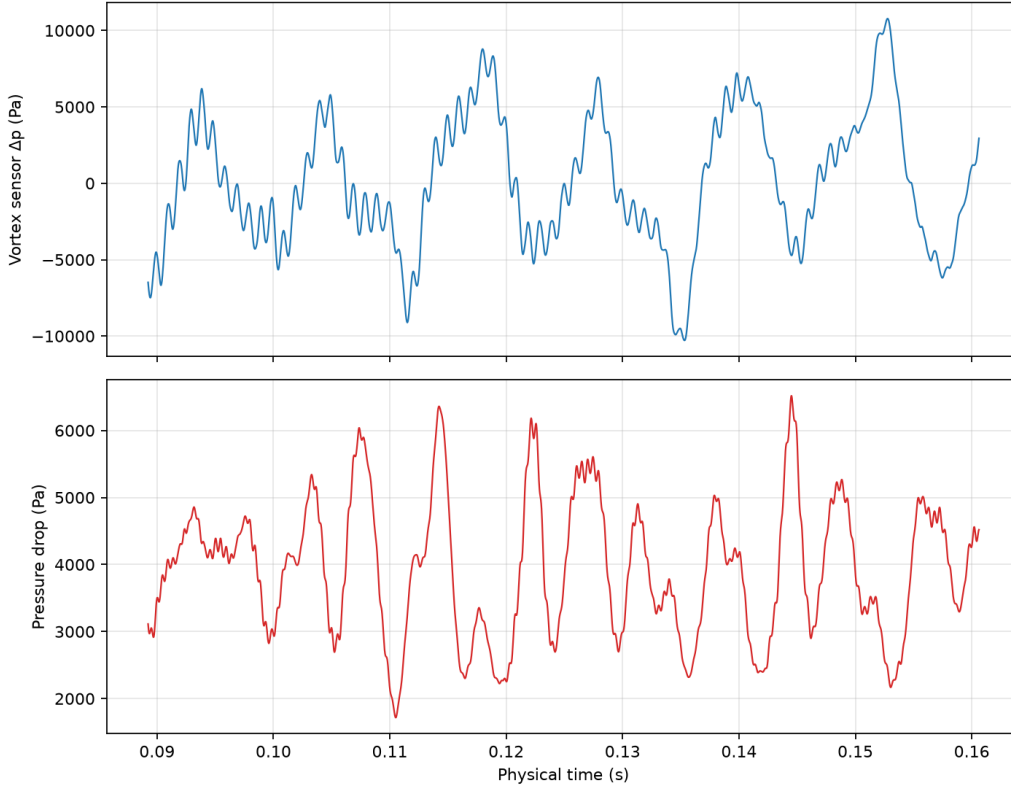


Figure 2: Wake-pressure difference and pressure-loss time series for the verified 39.189 mm case.

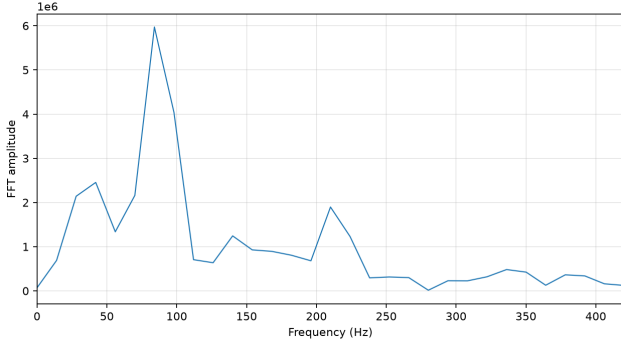


Figure 3: FFT spectrum of the verified wake-pressure signal.

The QR curve predicted $\hat{J}(39.189) = 0.1830$, while the fresh simulation produced $J_{LBM}(39.189) = 0.2337$. The absolute difference was 0.0507. The verified value was higher than every original sweep objective, so 39.189 mm is the best observed candidate. However, the prediction difference means the exact optimum is not yet established.

The nominal flow rate obtained from Eq. 1 is $0.07573 \text{ m}^3/\text{s}$ and the corresponding nominal mass flow rate is 2.0578 kg/s . These are reference physical pipe quantities. The optimization does not

attempt to increase the imposed flow rate; it attempts to improve measurement quality while controlling pressure loss.

4.6 Comparison with the Base Paper

The base paper selected 25 mm as part of a three-parameter geometry using D3Q19 MRT-LBM-LES at much higher Reynolds numbers. The present project predicts a promising region near 38–40 mm using a two-dimensional D2Q9 PyLBM model, different probes and a custom weighted objective. Therefore, the radius difference should not be interpreted as disproving the paper. It demonstrates that an optimum depends on model dimension, Reynolds number, geometry construction, measurement definitions and optimization criteria.

The paper used lift force, shedding behavior, symmetry and pressure loss. Our reduced implementation replaces direct surface lift measurement with an upper-minus-lower wake-pressure signal. This choice keeps the implementation manageable, but it is also a source of model discrepancy.

4.7 Limitations

The main limitations are:

- the flow model is two-dimensional and does not include the paper's full LBM-LES treatment;
- only five direct radii were used;

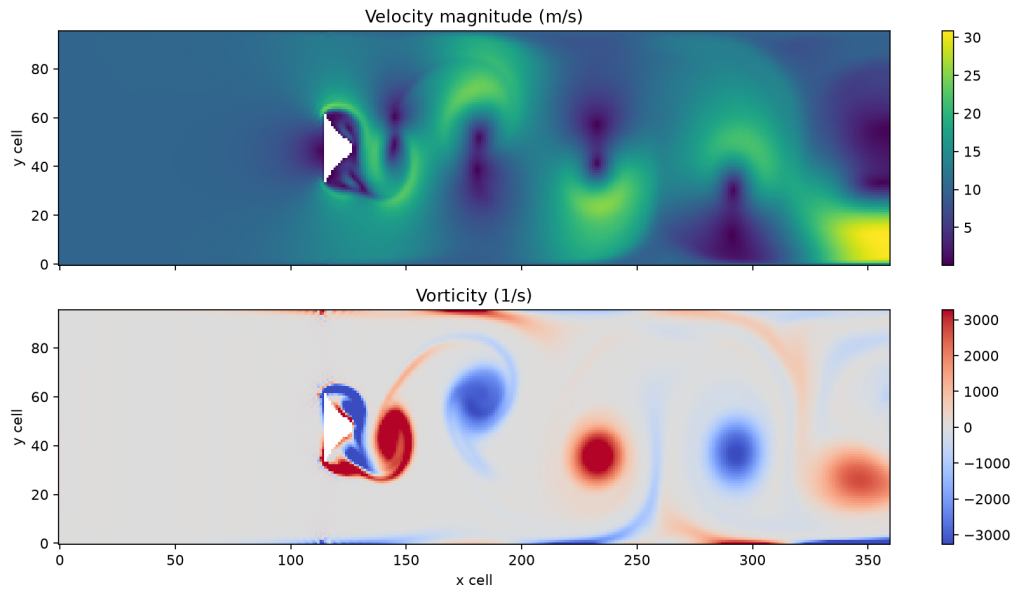


Figure 4: Velocity magnitude and vorticity for the verified 39.189 mm case.

- objective weights were selected for this project and were not calibrated experimentally;
- the FFT resolution is limited by the sampled duration;
- the surrogate error near the candidate is not negligible;
- no experimental measurements are available for validation;
- height, incoming angle and fillet radius were not re-optimized.

5 Conclusion and Future Work

This project implemented a complete numerical shape-study workflow using Python and PyLBM. Five side radii were simulated, and the resulting pressure and velocity fields were converted into frequency, Strouhal number, signal amplitude, pressure loss, symmetry and vorticity measurements. The work demonstrated course numerical methods including least-squares detrending, FFT, peak detection, trapezoidal and Simpson integration, numerical differentiation, polynomial interpolation, cubic splines, QR fitting, Golden-Section Search and Newton optimization.

Golden-Section Search predicted a radius of 39.189 mm on the QR surrogate. Newton's method was rejected because it found a minimum. A new PyLBM simulation at 39.189 mm passed all implemented quality checks and produced the highest directly observed objective. The correct conclusion is that 39.189 mm is the best verified candidate within the current 20–40 mm experiment, not a universal or mathematically proven optimum.

Future work should add direct cases at 37.5, 38.0, 38.5, 39.0, 39.2, 39.5 and 40.0 mm, repeat important cases to estimate numerical variability, conduct a grid-convergence study and examine sensitivity to the objective weights. A larger study could also optimize height, incoming angle and fillet radius, but a full multi-parameter design would require substantially more simulations. Finally, a three-dimensional LBM–LES model and experimental flowmeter data would provide stronger physical validation.

6 Github Repository

The public source-code repository is available at <https://github.com/FAHIM-ISHTIAK/CSE-402-Project>

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